Hotel Management System

SRS (Software requirement specification)

Introduction

1. Purpose of this document :- The purpose of this document is to outline the requirements and specifications for the development of a Hotel Management System. It will provide a clear understanding of the project objectives, scope and deliverables.

1.2 Scope of this Document : This document defines the overall working and main objectives of the Hotel Management System. It includes a description of the development cost and time required for the project.

General Descri.

1.3 Overview - Hotel Management system will cater to th needs of hotel staff and management, providing features such as room booking, guest profiles, inventory management, and financial reporting. It will be accessibl to users with varying levels of technical expertise.

3 Functional Requirements :

3.1 Reservation Management :

- Allow users to make room reservation online or through the front desk. Generate reservation confirmation and send notifications to guests.

3.2 Room Management:

- Assign rooms to guests based on availability and preferences
- Track room status (clean, occupied, vacant)
  send reservation ~~notifications~~ informations and notifications to ~~desk~~ guests.

3.3 Guest management

Maintain guest profiles with personal information preferences and booking history.
Facilitates Guests check in check out processe.

3.4 Billing and Invoicing
Generate accurate bills for room charges, additional sources, and taxes. Accept various payment methods and generate invoices for ~~co~~rporate clients.

4 Interface Requirements

4.1 User Interface:

- Intuitive and user friendly interface for hotel staff and guests.
- Accessible via web browsers, mobile devices and desktop ~~appliances~~ cations.

4.2 integration ~~with~~ Interface:
Integration with payment gateways for secure transactions
- Integration with 3rd party booking platforms for seamless reservation management

5) Performance requirements.

5.1 Response time:
The system should respond to user action within 2 seconds.

5.2 Scalability:
- Handle a minimum of 1000 concurrent users during peak hours.

5.3 Data Integrity -
- Ensure data consistency and accuracy across all modules.

6. Design constraints

6.1 Hardware limitations:
- The system should be compatible with standard hotel hardware (computer, printers, pos, terminals).

6.2 Software dependencies:
- Util ic a relational database management system (eg. MySQL) for data storage.
Use programming language and frameworks conducton to UML modelling (eg. java, springBoot)

7) Non functional attributes -

7.1 security:
- Implement robust authentication and authorisation mechanisms to protect sensitive dat

7.2 reliability
Ensure high availability & fault tolerance to minimise system downtime.

7.3 Scalability
Design the system to accomodate future growth and expansion.

7.4 Portability
- support multiple platforms and devices for user accessibility.

7.5 Usability
The system shall have user friendly interface with clear navigation

7.6 Reusability.
The system shall use modular code design to facilitate future enhancements and maintenance

7.7 Compatibility
The system shall be compatible with common web browsers (Chrome, Firefox, safari).

7.8 Data integrity.
The system shall ensure accurate and consistent data storage and retrieval.

8) Preliminary schedule and Budget - The development of the HMS is estimated to take months with budget of $10,000. This includes project planning, development, testing & deployment phases.

Preliminary Schedule

Assume the project is completed by a team of 3-4 students over 8 weeks.

S. No	Task	Duration
1.	Requirement gathering and analysis	week 1
2.	System design using UML diagrams	week 2
3.	Database design and setup	week 3
4.	Room booking and reservation module	week 4
5.	Customer management & check in / check out module	week 5
6.	Billing and payment module.	week 6
7.	System integration & testing	week 7
8.	Documentation & final deployment	week 8

2. Preliminary Budget

Resource	Estimated cost (₹)
1. Hardware & computer usage	5000
2. Software and development tools	0
3. Database setup	0
4. Internet & electricity	2000
5. Testing and debugging	1000
6. Documentation & printing	1000
7. Miscellaneous expenses	10,000
Total Estimated budget	10,000